

Viewpoint Suggestion of Feature-rich Photos Using Smartphone Photogrammetry and Keypoint Detection

Haruka Ono*
Tokyo City University

Koya Narumi†
Keio University

Yuki Uranishi‡
The University of Osaka

ABSTRACT

We propose a smartphone-based system that automatically suggests the suitable camera position and angle to a target object when we try to take a feature-rich photo of it. The system first captures a 3D mesh of the object using photogrammetry and virtually takes photos of the mesh from different positions and angles. Then, the system extracts and compares the keypoints of each virtual shot and outputs the one with the largest number of keypoints. We describe the system and demonstrate its effectiveness through examples.

Index Terms: Photogrammetry, Smartphone, Camera Viewpoint Suggestion

1 INTRODUCTION

The widespread adoption of smartphones has made photography an everyday activity, not only for professionals but also for general users. People try to take attractive photos of products they sell for online shopping services, selfies in front of beautiful scenery, and dishes they cook for social networking services.

However, taking high-quality, visually pleasing photos remains challenging due to complex factors such as camera distance, object poses, and scene composition, all of which require expertise for non-professional users. To solve this issue, several studies [2, 6] have proposed methods to improve photo composition. However, such methods are limited to objects with low angle dependency, i.e., distant landscapes that do not significantly change their appearance even when the user moves and rotates the camera. In other words, these methods cannot suggest a camera viewpoint when users can move around objects, as is common when taking photos of relatively small items, such as products, portraits, and landmarks.

We propose a system that automatically suggests the good camera position and angle as the user explores various viewpoints around an object with a smartphone. The system first captures a 3D mesh and *virtually* generates photos of the mesh from different positions and angles. The system then extracts and compares the keypoints of each virtual shot, and suggests the one with the largest number of keypoints.

The contributions of this paper include a smartphone-based system that suggests a feature-rich camera viewpoint for photographing a target object and an example-based evaluation demonstrating the effectiveness and limitations of the approach.

2 RELATED WORK

Some previous methods suggest cropped images of given photos for better appearance [3, 7, 13]. For example, SmartEye [6] suggested a cropped image based on a machine learning model called View Proposal Network (VPN) [9]. Another approach provides visual guidance to the user, for instance by visualizing instructions [1, 4, 11] or

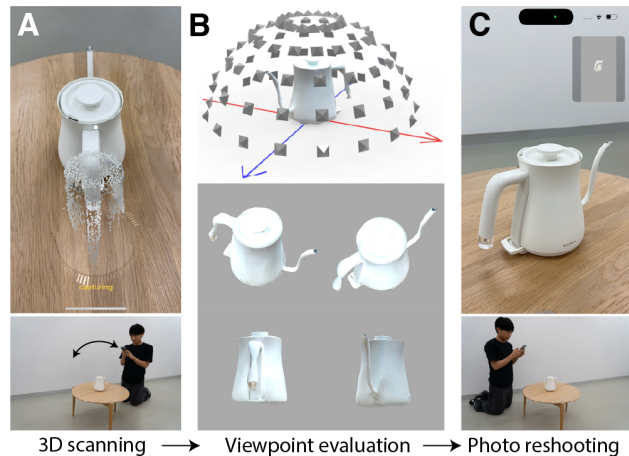


Figure 1: System overview. A: The user first captures a 3D mesh of a target object using photogrammetry. B: The system automatically generates *virtual* snapshots of the 3D mesh and evaluates them based on the keypoint detection. C: The user retakes a photo from the suggested viewpoint.

highlighting distracting elements in the background [2]. However, these approaches are based on 2D images and do not account for the subject's three-dimensional shape. Therefore, they cannot suggest better viewpoints when an object's appearance changes significantly with camera angle.

Other studies have explored enhancing the visual quality of images by considering the 3D geometry or depth information of the scene [8, 10, 12]. For example, Yamamoto et al. [12] proposed a system to optimize the lighting condition of a target with movable light bulbs attached to robot arms. However, this method requires a special setup and does not apply to daily situations.

In contrast to these previous approaches, our system focuses on providing real-time viewpoint recommendations for an angle-dependent target without special equipment.

3 SYSTEM DESIGN

The system consists of three main steps shown in Figure 1. First, during *3D scanning*, the user takes multiple images of the target object and reconstructs a 3D mesh by photogrammetry. Next, in *Viewpoint evaluation*, the system generates virtual renderings of the object from multiple viewpoints. Each rendered view is evaluated based on SIFT [5] keypoints. Finally, in *Photo reshooting*, the user moves to the suggested position, referring to the rendered image with the highest score, and retakes the photo.

3D Scanning To reconstruct a 3D object using photogrammetry, we used an iPhone 14 Pro and the Object Capture API in the Apple RealityKit framework. This API uses RGB images captured by the device camera and depth information obtained from the built-in LiDAR sensor, and outputs the reconstructed 3D mesh. Figure 1A shows a screenshot of our system and the user's scanning process.

*e-mail: g2331021@tcu.ac.jp

†e-mail: narumi@keio.jp

‡e-mail: yuki.uranishi.cmc@osaka-u.ac.jp

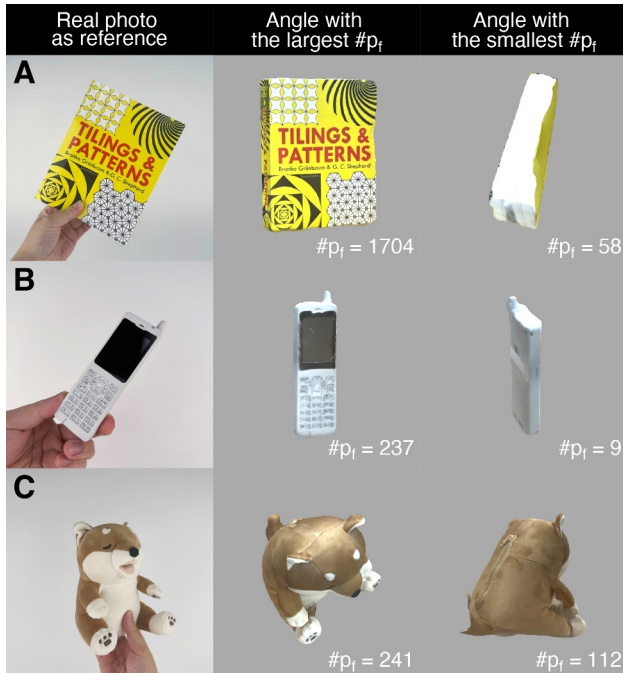


Figure 2: Evaluation results. Three columns show the real photo as reference, the virtual snapshots with the largest and smallest $\#p_f$. Three rows show a book (A), a codeless phone (B), and a puppy (C).

Viewpoint Evaluation Given the 3D mesh of the target object, the system takes approx. 90 virtual snapshots in the hemispherical coordinate system (θ, ϕ, r) . We sampled 5 points for θ and 18 points for ϕ . The distance was fixed at $r = 1\text{m}$ because our metric is scale-invariant. To assess the visual quality, we use the number of keypoints detected by Scale-Invariant Feature Transform (SIFT) [5] as a heuristic metric $\#p_f$. This approach assumes that photos with more features (e.g., corners, edges, and rich textures) contain more visual information, which may correlate with human judgment.

Photo Reshooting Finally, the system suggests the virtual snapshot with the highest $\#p_f$, which is shown on the top right of the smartphone screen (Figure 1C). The user refers to the virtual snapshot, moves to the appropriate position, and retakes a photo.

4 EVALUATION

Figure 2 shows the evaluation results of our system using three example objects. First, a book (Figure 2A) and a codeless phone (Figure 2B) demonstrate the feasibility of our keypoint-based heuristic, with a clear difference in $\#p_f$ in the largest and smallest cases. The virtual snapshots with the largest $\#p_f$ are promising, which capture the front side of the objects with more information. On the other hand, a puppy (Figure 2C) was relatively disappointing, with a smaller difference in the largest and smallest $\#p_f$. Likewise, the virtual snapshot with the largest $\#p_f$ is not satisfying, as it captures the top side of the puppy instead of the front.

We infer that Figure 2AB were successful because important angles have characters and components that contributed to the greater number of keypoints, while Figure 2C does not contain rich texture or color. Also, lighting conditions may have contributed to the larger number of keypoints on the top side of the puppy.

5 CONCLUSION

In this paper, we have proposed a smartphone-based system that automatically suggests the suitable camera position and angle to a

target object for a feature-rich photo using keypoint detection for viewpoint evaluation. According to the initial results, our method was found to be effective for specific scenarios. As future work, we will conduct user studies to investigate how well our keypoint-based metric aligns with human judgment, compare different keypoint detection methods or metrics, and evaluate the stability of our method through repeated tests. Additionally, we plan to implement an AR-based guidance system to help users retake a photo.

This work was supported by JST Science and Technology Challenge Program for Next Generation.

REFERENCES

- [1] J. L. E. O. Fried, J. Lu, J. Zhang, R. Mech, J. Echevarria, P. Hanrahan, and J. A. Landay. Adaptive photographic composition guidance. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, p. 1–13. Association for Computing Machinery, New York, NY, USA, 2020. 1
- [2] J. L., K. Y. Zhai, J. Echevarria, O. Fried, P. Hanrahan, and J. A. Landay. Dynamic guidance for decluttering photographic compositions. In *The 34th Annual ACM Symposium on User Interface Software and Technology*, UIST '21, p. 359–371. Association for Computing Machinery, New York, NY, USA, 2021. 1
- [3] D. Li, J. Zhang, K. Huang, and M.-H. Yang. Composing good shots by exploiting mutual relations. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4212–4221, 2020. doi: 10.1109/CVPR42600.2020.00427 1
- [4] J. Li, Y. She, F. Liu, C. Yu, X. Wang, and Y. Xu. Augmented reality based video shooting guidance for novice users. *Proc. ACM Hum.-Comput. Interact.*, 6(MHCI), Sept. 2022. doi: 10.1145/3546750 1
- [5] D. Lowe. Object recognition from local scale-invariant features. In *Proceedings of the Seventh IEEE International Conference on Computer Vision*, vol. 2, pp. 1150–1157 vol.2, 1999. doi: 10.1109/ICCV.1999.790410 1, 2
- [6] S. Ma, Z. Wei, F. Tian, X. Fan, J. Zhang, X. Shen, Z. Lin, J. Huang, R. M  ch, D. Samaras, and H. Wang. Smarteye: Assisting instant photo taking via integrating user preference with deep view proposal network. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, p. 1–12. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3290605.3300701 1
- [7] Y. Tu, L. Niu, W. Zhao, D. Cheng, and L. Zhang. Image cropping with composition and saliency aware aesthetic score map. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(07):12104–12111, Apr. 2020. doi: 10.1609/aaai.v34i07.6889 1
- [8] T. Uchida, Y. Kanamori, and Y. Endo. 3D view optimization for improving image aesthetics. In *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, pp. 1–5, 2025. 1
- [9] Z. Wei, J. Zhang, X. Shen, Z. Lin, R. Mech, M. Hoai, and D. Samaras. Good view hunting: Learning photo composition from dense view pairs. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5437–5446, 2018. 1
- [10] D. Xie, P. Hu, X. Sun, S. Pirk, J. Zhang, R. Mech, and A. E. Kaufman. Gait: Generating aesthetic indoor tours with deep reinforcement learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 7409–7419, October 2023. 1
- [11] Y. Xu, J. Ratcliff, J. Scovell, G. Speiginer, and R. Azuma. Real-time guidance camera interface to enhance photo aesthetic quality. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, CHI '15, p. 1183–1186. Association for Computing Machinery, New York, NY, USA, 2015. doi: 10.1145/2702123.2702418 1
- [12] K. Yamamoto, Y. Koyama, and Y. Ochiai. Photographic lighting design with photographer-in-the-loop bayesian optimization. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, UIST '22. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3526113.3545690 1
- [13] B. Zhang, L. Niu, X. Zhao, and L. Zhang. Human-centric image cropping with partition-aware and content-preserving features, 2022. 1